PROJECT BRIEFING

Contactless-to-Contact Fingerprint Recognition

What the prior work did, what we built, and what we achieved

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1. Why this document exists

This briefing aligns the team on three things: what the earlier work (VerifNet and the wider literature) actually did, what our project contributed, and how to describe the results honestly. It is written in plain English so that any teammate can read it once and then explain the project to the professor or in a viva without memorising jargon.

2. The problem, in plain English

A fingerprint system normally compares two contact images, both taken by pressing a finger on a scanner. In this project the two images come from different worlds: one is a photo taken with a phone camera (contactless), and the other is a scanner or live-scan image (contact).

The identity is the same, but the image language is different: lighting, focus, perspective distortion, skin deformation, ridge clarity and background all change. The task is to make the system say “these two images are the same finger” despite that gap. It is closer to matching a passport photo with a sketch than to matching two scans.

3. What the previous work did

3.1 VerifNet

VerifNet is the paper we started from. It uses a “committee of three experts”:

- **Wavelet features —** classic texture descriptors that capture ridge patterns.
- **A deep learning branch —** a Siamese CNN trained to pull images of the same finger together and push different fingers apart.
- **Minutiae matching —** the traditional method: locate ridge endings and bifurcations, then match their positions and angles.

Each expert produces a similarity score, and VerifNet fuses the three scores, optionally weighted by image quality. The argument is that the cues are complementary, so fusion is more robust than any single expert. This is a sound idea, and it is the baseline we reproduced.

3.2 The wider literature

Family	Representative methods	Strength	Weakness for this task
Classical minutiae	VeriFinger, NBIS, Minutiae Attention Network	Interpretable; strong on clean contact prints	Brittle on contactless images, where blur and distortion destroy minutiae
Deep embeddings	Siamese/contrastive CNNs, ArcFace, DeepPrint	Learns texture and tolerates distortion better	Does not by itself force the two modalities into a shared space
Fusion and transformer pipelines	VerifNet, Ridgeformer, IMPOSE, Multi-Siamese, MANet	Best raw accuracy in the literature	Heavy: several branches, GPU training, large external data; errors compound

Where the gap was. The gap we targeted follows directly from the last column: can we get strong cross-modal accuracy from a single, small, CPU-trainable embedding by explicitly forcing the two modalities to agree, instead of gluing several experts together after the fact?

4. What we did

4.1 Stage 1 — an honest baseline (the "v1" pipeline)

We first reproduced the VerifNet-style recipe under a strict protocol: a minutiae branch, a deep branch, and quality-aware score fusion. We also built ISE (Identity-aware Scattering Embedding): a fixed 2-D scattering transform, which is a mathematically defined texture descriptor with no learned first layer, feeding a small CNN trained with identity supervision.

The result was informative. Fusion helped, but the single ISE embedding was already far better than minutiae or the plain VerifNet-style deep branch. That told us the real problem was not which experts to fuse, but how to align the two modalities inside one embedding.

4.2 Stage 2 — our method: CMRA

CMRA stands for Cross-Modal Relational Alignment. It is one shared embedding network with three ideas layered on top.

1. **One translator, not three experts.** Both contactless and contact images pass through the same small CNN and emerge as a 128-dimensional vector. At test time we compare two vectors with cosine similarity; there is no fusion step at all.
2. **Identity-balanced "PK" batches.** Each training batch is a small classroom containing 16 identities × 2 contactless + 2 contact images (64 images). Every identity appears in both modalities in every step, so the model is constantly forced to compare the two image languages directly.
3. **Two consistency losses.** Two consistency terms do the real work. The prototype loss makes the average embedding of an identity's contactless images equal the average of its contact images: the same finger should have one centre of gravity whichever camera took the picture. The relational loss makes both modalities agree on who is close to whom: if finger A is close to finger B in the photo world, it should also be close to B in the scanner world. It is implemented as a temperature-scaled symmetric KL divergence between the two similarity matrices.
4. **Risk-controlled abstention.** Pairs whose score is close to the decision threshold can be rejected as "uncertain" instead of being forced into accept or reject. In a security setting the system can say "send this one to a human", trading coverage for a lower error rate.

$$\textbf{Training objective:}\quad L = L_{ArcFace} + 0.5 \cdot L_{prototype} + 0.5 \cdot L_{relational}$$

4.3 Why this is genuinely different

VerifNet combines different feature types after the fact. CMRA uses one feature type and changes the training objective so the two modalities are aligned from the start. The comparison is therefore not “our fusion versus their fusion”; it is “alignment inside the model versus fusion outside the model”.

5. What we achieved

All numbers below are on the identity-disjoint HKPolyU CL2CB test set: 10,728 pairs (5,364 genuine + 5,364 impostor), with no identity overlap between training, validation and test. This is deliberately harder and more honest than random image splits.

5.1 Headline comparison

System	EER % (lower)	TAR @ FAR 1% (higher)	TAR @ FAR 0.1% (higher)
Minutiae only	39.29	6.41	1.96
VerifNet-style deep branch	36.11	3.56	0.15
v1 fixed fusion (deep + minutiae)	32.83	8.35	2.16
ISE (single embedding, v1)	19.56	28.24	4.05
v1 best quality-aware fusion	18.16	32.94	12.19
CMRA full, seed 7	16.80	29.55	7.20
CMRA full, seed 8 (best seed)	16.06	30.93	11.09

5.2 What the ablation shows

At seed 7, ArcFace with PK batches alone reaches 19.41% EER — essentially the ISE level. Adding the prototype loss gives 18.23%; adding the relational loss instead gives 17.74%; adding both gives 16.80%. The gain therefore comes from the alignment losses, not from fusing more branches.

Configuration (seed 7)	EER %	TAR @ 1%	TAR @ 0.1%
ISE (v1 single embedding)	19.56	28.24	4.05
CMRA ArcFace + PK only	19.41	24.33	4.34
+ prototype loss	18.23	27.44	5.87
+ relational loss	17.74	28.93	6.99
+ both (full CMRA)	16.80	29.55	7.20

5.3 Cross-session and abstention

On the 48 identities that appear in both sessions, CMRA reaches 14.89% cross-session EER, against 18.50% for the v1 quality-aware fusion, 18.89% for ISE and 37.66% for minutiae. CMRA is therefore better when the two images come from different sessions — a good sign that it learned modality-invariant structure rather than session-specific cues.

With risk-controlled abstention (seed 8), the error rate falls from 16.04% at full coverage to 9.94% at 80% coverage and 3.88% at 50% coverage. The area under the risk–coverage curve is 0.056 for CMRA versus 0.084 for ISE, so its confidence ranking is also better.

5.4 Honest caveats

State these before the professor asks

Seed variance. Full CMRA across seeds 7, 8, 9 and 10 gives 16.80, 16.06, 18.96 and 20.15% EER. The stable seeds average about 17.3%; seed 9 is an outlier. We report a range, not only the best run.

Low-FAR tail. The v1 fusion still wins at FAR = 1% and 0.1% (32.94 / 12.19 versus 30.93 / 11.09 for the best CMRA seed). CMRA's win is at EER, with a much simpler pipeline.

No state-of-the-art claim. Published 2–4% EER methods use transformers, external data or generative augmentation. We deliberately stayed CPU-trainable (about 0.13 M parameters) and reproducible.

Scope. One dataset, with same-session genuine pairs as the primary test and the cross-session study as a supplementary check.

6. The 60-second version to say out loud

"VerifNet and similar methods fuse three separate experts — wavelets, a deep network and minutiae — to bridge the contactless-to-contact gap. That works, but it is a heavy pipeline and the branches are still trained separately. We asked whether a single small embedding could do better if we explicitly force the two modalities to agree during training. Our method, CMRA, uses one shared CNN on scattering features, identity-balanced batches that always contain both modalities, and two consistency losses: a prototype loss that aligns each identity's average position across modalities, and a relational loss that makes both modalities agree on who is close to whom. On the identity-disjoint HKPolyU protocol this reduced EER from 19.56% for a single embedding and 18.16% for the best three-branch fusion to 16.80% at seed 7 and 16.06% at seed 8, with a stable-seed average around 17.3%. Across sessions it improves further, from 18.50% to 14.89%. The ablation shows the gain comes from the alignment losses, not from extra fusion. The three-branch fusion still has a slightly better low-FAR tail, so our honest claim is comparable-or-better EER with a simpler, genuinely different pipeline — not state of the art. We also added risk-controlled abstention, which cuts the error to about 3.9% if the system is allowed to reject half of the uncertain pairs."

One-sentence version: Instead of fusing three experts after the fact, we trained one small embedding so that the contactless and contact views of the same finger occupy the same place — and that alone beats the fusion baseline on EER.

7. Likely follow-up questions

Why is the EER better but TAR at 1% not? EER summarises the whole operating range; TAR at 1% is one strict point. The v1 fusion calibrates better in the extreme tail because it has three independent cues. It is a different trade-off, not a contradiction.

Is 16% good? On this identity-disjoint, same-session protocol with a CPU-only model, it is a large improvement over the reproduced baselines. Published 2–4% numbers use different splits, GPU training, external data or transformers, so they are not directly comparable.

What is actually new? The training objective: prototype plus relational consistency on identity-balanced PK batches, applied to contactless-to-contact matching. VerifNet fuses features; we align representations.

Why scattering instead of a pretrained CNN? It is fixed, small and reproducible without pretraining data. It keeps the comparison fair and the pipeline CPU-friendly.

What would you do next? Add a transformer or attention head, evaluate on a second dataset, and calibrate the abstention threshold to a stated security policy.

8. Where everything lives

Every number in this briefing is backed by a file in the repository.

Artefact	Location
Ablation table, confidence intervals	results/cmra_metrics.csv
Cross-session comparison	results/cmra_cross_session.csv
Risk–coverage data and figure	results/cmra_risk_coverage.csv, cmra_risk_coverage.png
DET curve	results/cmra_det.png
Paired tests and summary	results/cmra_summary.json
Per-run checkpoints and scores	results/cmra_full_s{7,8,9,10}/
Full report (Word, PDF, Overleaf source)	report/
Final slide deck	slides/
Reproduction commands	README.md

Repository housekeeping: scratch scripts, debug images, smoke-test runs, training logs, caches and the superseded proposal deck have been removed. The core source, all metrics and scores, the report, the slides and the reference papers are kept. The dataset and environment folders (src/data and venv, about 14.7 GB together) are also still in place so that results can be regenerated; remove them only if the team agrees.